Sébastien VALAT
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Date of birth: 25 january 1985
(33 year old)
Nationality: French
Marital status: Single
French driving licence

Researcher in computer science

Work experience

Since 07/2015 — CERN / LHCB
Fellow on developpement studies for next 40 Tb/s Data Acquisition System. Was working on InfinBand and Omni-Path fabrics with target : 500 nodes.

12/2013 - 12/2014 — Exascale Computing Research, UVSQ
Research engineer on a dynamic memory allocation profiling tool for parallel applications. It generates profiles of malloc usage mapped on call stacks.

11/2010 - 11/2013 — CEA,DAM,DIF
PhD. on memory managment for HPC : parallel and NUMA memory allocator, kernel aspects and cache effects. Usage of supercomputers Tera 100 and Curie. Part of developpement team for MPC (MultiProcessor Computing) runtime.

2009 (5 months) — CEA,DAM,DIF
Master 2 internship : Linux kernel module developpement to add support for page coloring and evaluate impact on caches. I observed unexpected effects I studied in my PhD. thesis.

2008 (4 months) — Laboratoire d'Annecy-le-vieux de Physique des Particules
Master 2 internship on LHCb to study the CP symmetry violation channel : $Bs \rightarrow J/\Psi\eta$. Event reconstruction on CERN grid of simuated traces.

2007 (6 weeks) — Laboratoire d'Annecy-le-vieux de Physique des Particules
Master 1 internship on OPERA detector to reconstruct neutrinos (ν_μ, ν_τ) decays with 2 verticies.

Training

2013 (2 weeks)
- ▷ CEA/INRIA/EDF summer school : programming heteregoneous parallel architectures (OpenACC, CUDA, StarPU, C++AMP).

2012 (1 week)
- ▷ CNRS summer school: Hybride programming : a step to many-cores.
- ▷ CNRS summer school: Methodology and tools for optimization.

2011 (1 week)
- ▷ CEA internal advanced training for modeling and simulation methods applied to various CEA domains.

Education

2010 - 2014 — Commissariat à l'Énergie Atomique / university of Versailles
PhD. in computer science : memory managment for HPC.

2008 - 2009 — University of Savoie
Master degree in computer science and cooperative systems. *Ranked 1st.*

2007 - 2008 — Joseph Fourier University
Master degree in subatomic and astroparticles physics.

2003 - 2007 — University of Savoie
1st year of master degree in sciences and technologies oriented to physics.

2003 — High school diploma in science specialized to physics.

Languages

French (natal language). English (fluent).

Teaching / supervising

2014	▷ Lab for system administration (3rd year of ingineer school ISTY/UVSQ).
2012	▷ Lab onto MPC for CNRS summer school on hybrid programming.
2011 – 2013	▷ Co-supervise an ingineer apprentice on MPC : Jenkins, test suite scheduling up to 3000 cores, ports of memory allocator onto Windows.
2010 – 2013	▷ Lab for hardware architectures (2nd year of ingineer school ISTY/UVSQ). ▷ Lab for parallel MPI/OpenMP optimization (master degree MIHPS/UVSQ).

Skills

Common langages :	C/C++, Shell, xml/xslt/dtd, php/javascript/html/css, NodeJS, Python, Java.
Known languages :	Perl, C#, SQL, Assembler, Ruby.
Parallelism :	MPI, Pthread, OpenMP, CUDA (limited).
Librairies :	Qt, Gtest, libxml, Root, Hwloc, OpenGL, libfabric.
Tools :	Autotools, Cmake, Maven, Trac, Svn, Git, Gdb, Jenkins, Qemu, Slurm.
Other :	System and Linux kernel developpement.
Technologies :	InfiniBand, Omni-Path.

Projects

2017	**NUMAPROF** A NUMA profiling tool based on pintool, using similar approach than MALT. The idea is to annotate sources linux with remote/local accesses.
2015 - 2017	**DAQPIPE** (DAQ Protcol Independent Performance Evaluator). Developpement of a benchmarking applicaton for future DAQ of LHCb supporting multi-networkds like InfiniBand, Omni-Path, ethernet and RapidIO.
2013 - 2014	**MALT** (A Malloc Tracker): Developpement of a memory profiling tools to C/C++/Fortran applications by annotating the source code.
2010 - 2013	**MPC** (MultiProcessor Computing) : runtime support for NUMA clusters developped by CEA (MPI, OpenMP...). In charge of the *memory allocator*. Involed in validation and technical management of the project.
2012	**LBM** (Lattice Boltzmann Method) : simulation of Karman vortices in C/MPI/OMP built for parallel optimization labs given to master MIHPS.

Personal projects

Since 2012	**CMR** (Cartesian Mesh Runtime) : a specialized language based on Latex with a C++ library for cartesian meshes simulation in MPI + OpenMP.
2013	**HTOPML** : tools to track behavior of a remote living code with a web GUI (processor usage, thread bindings, memory usage...).
2010	**SVUNITTEST** : C++ unit test library similar to cppunit/gtest.
2008	**CORTEX** : cluster based on MOSIX and XEN for C-Net (4 nodes quad-cores).
2007	**Emulator and assembler** for microship Motorola 6809.
2006	**TSA** : webservice to administrate Trac/SVN repositories for C-Net.

Volunteering

2011 – 2014	Member of a local diving club.
2004 – 2010	Member of the university IT association : C-Net. ▷ *2006 - President.* ▷ *2008 - Treasurer.* ▷ *Network addmin.* ▷ *C++ tutorials.*
2000 – 2010	Member of a local association organizing a rural film festival.

Hobbies

Scuba diving (N3), swimming, apnea, movie theatre, scientific culture, IT developpement, biking, skiing.

Publications

2017 (Vancouver) — **MALT, A Malloc Tracker**. *In proceedings of ACM SIGPLAN Workshop on Software Engineering for Parallel Systems (SEPS). DOI: 10.1145/3141865.3141867.*
▷ *S. Valat, A. R. Sharif and W. Jalby.*

2017 (Padua) — **An Evaluation of 100-Gb/s LAN Networks for the LHCb DAQ Upgrade**. *IEEE Transactions on Nuclear Science, vol. 64, no. 6, June 2017 (Real Time Conference 2016). DOI: 10.1109/TNS.2017.2687124.*
▷ *S. Valat, B. Voneki, N. Neufeld, J. Machen, R. Schwemmer and D. H. Cámpora Pérez.*

2017 (Padua) — **Exploring RapidIO Technology Within a DAQ System Event Building Network**. *IEEE Transactions on Nuclear Science, vol. 64, no. 6 (Real Time Conference 2016). DOI: 10.1109/TNS.2017.2734564.*
▷ *S. Baymani, K. Alexopoulos and S. Valat.*

2017 (Padua) — **Large-Scale DAQ Tests for the LHCb Upgrade**. *IEEE Transactions on Nuclear Science, vol. 64, no. 6, June 2017 (Real Time Conference 2016). DOI: 10.1109/TNS.2017.2693422.*
▷ *M. Manzali, A. Flababella, F. Giacomino, U. Marconi, N. Neufeld, S. Valat, B. Voneki.*

2013 (Seattle) — **Introducing Kernel-Level Page Reuse for High Performance Computing**. *In proceedings of ACM SIGPLAN Workshop on Memory Systems Performance and Correctness (MSPC). DOI: 10.1145/2492408.2492414.*
▷ *S. Valat , M. Pérache et W. Jalby.*

2010 (Stuttgart) — **A Decremental Analysis Tool for a Fine-Grained Bottleneck Detection**. *In proceedings of Parallel Tools Workshop.*
▷ *S. Koliai , S. Valat , T. Moseley , J. T. Acquaviva, et W. Jalby.*

Seminar and talks

10/2017 — **MALT, A Malloc Tracker**.
▷ *SEPS 2017 (Vancouver.*

06/2017 — **DAQ, Filtering Data from 1 Pb/s to 600 MB/s**.
▷ *CERN OpenLab summer student lecture.*

06/2017 — **Studying a 40 Tb/s data acquisition system for the LHCb experiment**.
▷ *InHP@CT teratech meeting.*

02/2016 — **Preparing LHCb event building at 4 TB/s**.
▷ *Open Fabric Allicance Workshop (Monterey, US.*

07/2014 — **Contribution à l'amélioration des méthodes d'optimisation de la gestion de la mémoire dans le cadre du Calcul Haute Performance**.
▷ *PhD. defense at university of Versailles.*

07/2013 — **Introducing Kernel-Level Page Reuse for High Performance Computing**.
▷ *In Workshop on Memory Systems Performance and Correctness (MSPC), Seattle.*

06/2013 — **Contribution à l'amélioration des méthodes d'optimisation de la gestion de la mémoire dans le cadre du Calcul Haute Performance.**.
▷ *CEA/DAM PhD. student day.*

10/2012 — **MPC labs**.
▷ *Labs at IN2P3 summer school "Ecole programmation hybride, une étape vers le many-coeurs ?", Autrans.*

02/2012 — **OS performance impact**.
▷ *Seminar at IN2P3 summer school "Méthodologie et outils d'optimisation en développement logiciel", Fréjus.*